ENTERPRISE WEB APPLICATION DEVELOPMENT

# CampusHub API Unit Testing Lab

Student handout / Spring Boot course controller / 45 minutes

**Goal.** Write and run five JUnit tests for the course API controller. Use the in-memory CampusHub version from the REST practical guide. Call controller methods directly, with Spring Boot and MySQL stopped.

## 1. Set up the test class

In the backend `pom.xml`, confirm that `spring-boot-starter-test` is present with `test` scope. Add it if needed and reload Maven.

Create `CourseControllerTest.java` under `src/test/java` in the controller package. Use one `@Test` method per case and a fresh `CourseController` in `@BeforeEach`. For tests 03 and 04, create the input with `new CourseInput(code, title, credits)`.

## 2. Implement these five test cases

| Test case | Call and input | Expected result |
|---|---|---|
| **01** Find an existing course | `getCourseById(101L)` | **200 OK.** The returned course has ID 101 and code IS-401. |
| **02** Look up a missing course | `getCourseById(9999L)` | **404 Not Found.** The response body is empty. |
| **03** Create a valid course | `createCourse(input)`<br>Input has code IS-404, title Software Testing, and 3 credits. | **201 Created.** The returned course has code IS-404 and 3 credits. |
| **04** Reject zero credits | `createCourse(input)`<br>Input has code IS-404, title Software Testing, and 0 credits. | **400 Bad Request.** The message explains that credits must be positive. |
| **05** Delete an existing course | `deleteCourse(103L)` | **204 No Content.** Looking up course 103 afterward returns 404. |

For each test, arrange the input, call the method, then assert the result. Use `getStatusCode().value()` to read the status and `getBody()` to inspect the returned value.

## 3. Run the tests and submit a short report

Open a terminal beside `pom.xml`. Run one of these commands.

```
Windows (PowerShell) .\mvnw.cmd -Dtest=CourseControllerTest test
macOS / Linux ./mvnw -Dtest=CourseControllerTest test
```

If there is no Maven wrapper, use `mvn -Dtest=CourseControllerTest test`. You can also run the entire test class from your IDE.

**Target.** Five tests run with zero failures, errors, or skipped tests. If a test fails, read the assertion output, fix the cause, and rerun the class.

**Submit one PDF report** with your name and student ID, a five-row results table using the headings below, and a screenshot of the final Maven or IDE test summary.

| Test ID | Expected result | Actual result | Pass / Fail |
|---|---|---|---|

Project reference: CampusHub REST Fundamentals Practical Guide, pp. 9-12. Reading: Sharma, Modern API Development with Spring and Spring Boot, Ch. 8.